NAIEVE STRING MATCHING

Naive string matching algorithm:-

It is basic string matching algorithm. The naive string matching algorithm compares characters one by one between the pattern and each substring of the text of the same length. The time complexity of the naive string matching algorithm is **O (n-m+1)m**.

		1	2	3	4	5	6	7	
T	=	a	c	c	a	b	c	d	[n = 7]
						1	2	3	
P	=					a	b	c	
									[m = 3]

Maximum Shift would be n-m 7-3=4

Shift 0= only a matches and mismatch occur at 2nd char so increment will occur in loop.

Shift 1 = mismatch occurs at 1st char so increment will occur in loop.

Shift 2 = mismatch occurs at 1st char so increment will occur in loop.

Shift 3= all characters match so loop will do increment.

Shift 4 = mismatch occurs at 1st loop will terminate.

Naive String Matching (T,P)

1. N = Length[T] -----1
2. M= Length [p] -----1
3. for (sh = 0; sh < n-m; s++)---n-m+1
4. if P[M] = T[Sh…………]
5. Then print "Pattern occurs at-----"

Complexity is O(n-m+1)m